

HUMAN-COMPUTER INTERACTION (CSSE4153)

Comprehensive Final Term Study Guide & Solved Past Papers (Spring/Fall)

Aligned with Syllabus Weeks 6 to 11 | UCP Midterm & Final Term Exam Patterns

Course: CSSE4153 - HCI	Target Exam: Final Term (40% Weightage)
Instructor: Dr. Noman Nazar	Syllabus Scope: Weeks 6, 7, 8, 9, 10, 11
Exam Format: MCQs, Short-Answer & Scenario Questions	Key Included: Full Past Paper Solutions & Question Bank

1. Prototyping: Lo-Fi, Hi-Fi & Storyboarding

A prototype is an early sample, model, or release of a product built to test a concept or process. In Human-Centered Design (HCD), prototyping allows teams to explore ideas and gather user feedback before investing in full engineering.

Dimension	Low-Fidelity (Lo-Fi)	High-Fidelity (Hi-Fi)
Definition & Medium	Paper sketches, 8-up storyboards, post-it notes, cardboard mockups.	Interactive digital prototypes (Figma, Adobe XD, HTML/CSS).
Cost & Velocity	Quick, cheap, disposable, and easily modified in minutes.	Expensive, time-consuming to build, harder to change.
Primary Purpose	Gathering requirements, testing big-picture structure & user flow.	User testing, look & feel, living specification for developers.
Why Early Lo-Fi is Better	1. Prevents distraction: Users focus on function rather than font/color. 2. Prevents over-attachment: Developers don't cling to code. 3. Encourages exploration: Easy to throw away.	Useful later in the design cycle once core interaction flows are validated.

Throwaway vs. Evolutionary Prototyping:

- **Throwaway (Rapid) Prototyping:** Prototypes built solely to explore requirements and gather early feedback, then completely discarded. Never forms part of the final codebase.
- **Evolutionary Prototyping:** An initial prototype is presented to users, refined based on feedback, and incrementally evolved stage-by-stage until it becomes the final delivered software system.

8-Up Storyboarding Protocol (Week 6/7 Activity):

- Fold a blank paper 3 times to create an 8-panel grid.
- Focus strictly on *problem descriptions* (user pain/irritation), NOT jump to solutions.
- Use proper nouns (e.g., 'Bilal', 'Sara') and capture explicit emotional states (e.g., frustrated, anxious, defeated).
- Include **What? How? Why?** observation captions for each frame.

2. Learnability & The Human Information Processing System

Learnability measures how easy it is for users to accomplish basic tasks the first time they encounter an interface. People learn primarily by **doing** (goal-directed exploration), not by reading manuals or taking training classes.

The Model Human Processor (MHP) & Cycle Times:

MHP models human cognitive architecture via three interacting processors:

Processor	Cycle Time	Key Function & Exam Relevance
Perceptual Processor	$T_p \approx 100$ ms (range: 50-200 ms)	Sensory stimuli processing. <i>Perceptual Fusion</i> : Stimuli within 100 ms are merged (e.g., video frame rates). Faster for strong stimuli.
Cognitive Processor	$T_c \approx 70$ ms (range: 25-170 ms)	Fetches information from Working Memory and makes strategy decisions. Faster when heavily loaded or practiced.
Motor Processor	$T_m \approx 70$ ms (range: 30-100 ms)	Executes physical muscular movement. Open-loop control = 70 ms; Closed-loop feedback cycle = $T_p + T_c + T_m \approx 240$ ms.

Memory Hierarchy & Chunking:

- **Sensory Memory**: Buffer holding raw visual/auditory signals for 200–500 ms.
- **Working Memory (WM)**: Short-term store holding 7 ± 2 chunks (or 4–5 items) for ~10 seconds without rehearsal. Maintenance rehearsal prevents decay.
- **Long-Term Memory (LTM)**: Infinite capacity. Elaborative rehearsal (connecting new info to existing mental models) permanently transfers chunks to LTM.
- **Chunking**: Grouping individual units into familiar meaningful patterns (e.g., grouping phone numbers 0300-1234567 or TLAs like IBM, FBI vs random letters MWBCRALOABIMBFI).

Recognition vs. Recall:

- **Recognition (Knowledge in the World)**: Remembering with the help of visible cues (GUI menus, icons, radio buttons). Requires minimal cognitive effort.
- **Recall (Knowledge in the Head)**: Retrieving facts from memory with no external help (Command-Line Interface parameters, blank text fields). Recognition is far easier and less error-prone.

3. Fundamental Design Principles (Norman's Framework)

Principle	Definition	UI / Real-World Example & Application
Affordance	Perceived and actual properties of an object that determine how it can be used.	3D raised buttons afford pushing; sliders afford dragging. <i>Perceived affordance</i> : Visual cues telling the user what to do.
Mapping	Relationship between controls and their actual effects on the system.	Physical stove burners matching knob layout; scrollbar moving viewport down when dragged down.
Feedback	Immediate visual, audio, or haptic reaction to user actions.	Button depressed state (<0.1s), progress bars (>10s). Low-level (click sound) vs High-level (page loaded).
Visibility	Making relevant options and system state visible to the user.	Breadcrumbs (Travel > Guides), highlighted active tabs, visible system status indicators.
Constraints	Restricting user interactions to prevent illegal inputs and errors.	Disabling past dates in calendar pickers; dropdown menus instead of free-text boxes for country entry.
Consistency	Principle of least surprise: similar things look/act the same.	Internal : Same button layout across screens. External : Following platform guidelines (Ctrl+C for Copy). Metaphorical : Trash can icon.

4. Efficiency: Fitts's Law, Steering Law & KLM-GOMS

Efficiency measures how quickly an expert user can complete routine tasks once the system is learned.

Fitts's Law (Pointing Model):

Predicts movement time (MT) to acquire a target based on distance (A = Amplitude) and size (W = Width):

Formula: $ID = \log_2(2A / W + c)$ | $MT = a + b \times ID$

- *ID (Index of Difficulty)*: Measured in bits.
- *a*: Reaction time / start-up time.
- *b*: Device pointing constant.
- *c*: Constant (0, 0.5, or 1).
- **Implications:** Make targets bigger, place controls closer together. **Edges & Magic Corners:** Screen edges have infinite width ($W = \infty$), making target acquisition fastest (e.g., Macintosh top menu bar is faster than Windows floating menu bar).

Steering Law (Tunneling / Drawing):

Predicts time (T) to navigate a cursor through a constrained tunnel of length D and width S:

Formula: $T = a + b (D / S)$

- *Core Difference:* Steering difficulty is **linear**, whereas pointing difficulty is **logarithmic**. Steering is significantly harder; this is why cascading submenus are poor UI design.

Keystroke Level Model (KLM-GOMS):

Calculates expert execution time for error-free routine tasks by summing operator execution times:

Operator	Standard Time	Description & Rules
K (Keystroke)	0.28 s	Pressing a single key (average typist = 0.28s; best = 0.08s; worst = 1.2s).
B (Mouse Button)	0.1 s	Press or release mouse button. Full Click = Press + Release = $2B = 0.2$ s.
P (Pointing)	1.1 s	Moving mouse cursor to a target on screen (based on Fitts's Law).
H (Homing)	0.4 s	Moving hand between keyboard and mouse (or vice versa).
D (Drawing)	1.1 s	Drawing straight line segments using pointer.
M (Mental)	1.2 s	Mental preparation. Rules to place M: Initiating a task, making a strategy decision, retrieving a chunk from memory, finding something on screen, verifying results.

5. Design Guidelines & Standards

Shneiderman's 8 Golden Rules:

1. Strive for consistency.
2. Enable frequent users to use shortcuts (accelerators).
3. Offer informative feedback.
4. Design dialogs to yield closure.
5. Offer error prevention and simple error handling.
6. Permit easy reversal of actions (Undo).
7. Support internal locus of control (make user feel in control).
8. Reduce short-term memory load.

Norman's 7 Principles:

1. Use both knowledge in the world and knowledge in the head.
2. Simplify task structure.
3. Make things visible (bridge Gulfs of Execution & Evaluation).
4. Get mappings right.
5. Exploit power of constraints.
6. Design for error.

7. When all else fails, standardize.

6. Perceptual Organization & Gestalt Laws

Explains how the human visual processing system automatically organizes low-level visual stimuli into structured wholes.

Pre-Attentive Processing vs. Feature Conjunction:

- **Pre-Attentive Processing:** Visual properties processed instantly (<200-250 ms) without conscious focus (e.g., color hue, line orientation, length, size, curvature).
- **Feature Conjunction:** When targets differ from distractors by a combination of two features (e.g., finding a red circle among red squares and blue circles), pre-attentive processing fails, requiring slow sequential visual search.

Gestalt Law	Visual Principle	UI Design Application
Proximity	Elements close to each other are perceived as a related group.	Grouping related form fields close together; separating sections with whitespace.
Similarity	Elements sharing visual attributes (color, shape) appear related.	All primary buttons having the same color; links styled consistently.
Continuity	Eyes follow continuous lines, curves, and visual paths.	Aligning form labels in vertical columns; stepped wizard progress bars.
Closure	Brain fills in missing gaps to perceive incomplete shapes as whole.	Effective use of white space; icons with open shapes (e.g., panda logo).
Figure-Ground	Eye separates dominant shape (figure) from background (ground).	Modal popup dialogs dimming background screen; high contrast text.
Common Fate	Elements moving together in the same direction belong together.	Selecting multiple desktop icons and dragging them together.

7. Evaluation Methods & Nielsen's Heuristics

Evaluation assesses the usability and acceptability of interactive systems.

4 Purpose Types & Classifications:

- **Exploratory:** Before design. Evaluates user needs and existing habits.
- **Predictive:** During design. Discounted inspection (KLM, Cognitive Walkthrough) without testing real users.
- **Formative:** During development. Informs design improvements; collects *qualitative process data* (why things went wrong).
- **Summative:** After completion. Assesses overall quality; collects *quantitative bottom-line data* (task time, completion rate, error count).

Nielsen's 10 Usability Heuristics & Severity Scale:

Severity Scale (0–4):

0 = Not a problem | 1 = Cosmetic problem only | 2 = Minor usability problem (low priority) | 3 = Major usability problem (high priority) | 4 = Usability catastrophe (imperative to fix before release).

Note: Single evaluator rating is unreliable; average of **3 to 5 evaluators** catches ~75–80% of usability problems.

Cognitive Walkthrough (Exploratory Learning Evaluation):

Expert evaluates a prototype step-by-step by answering 4 core questions at each user action:

1. Will users see that the action is available? (Visibility)
2. Will users know that the action is the one they need? (Recognition)
3. Is the effect of the action the same as the user's goal at that point?
4. After the action is taken, will users understand the feedback they get?

8. Solved Past Papers & Complete Question Bank

This section contains complete, exam-ready solutions for ALL short-answer questions, scenario case studies, quantitative KLM/Fitts's Law problems, and MCQs extracted directly from UCP HCI past exam papers (Spring/Fall 2021–2024).

Part A: Short Answer Questions (Complete Solutions)

Q1: What is a mile-high menu? Explain with an example.

A mile-high menu relies on Fitts's Law, where targets located at the physical edge of a computer display have an effectively infinite target width ($W = \infty$). Because the mouse cursor cannot move past the screen boundary, users can acquire the target at maximum speed without overshooting. Example: The Macintosh top menu bar is a mile-high menu because it resides at the top pixel boundary of the screen, making it faster to click than floating Windows menu bars.

Q2: Explain mapping with the help of an example.

Mapping is the relationship between interface controls and their physical or conceptual effects on the system. Natural mapping exploits spatial or physical analogies to make controls self-explanatory. Example: On a kitchen stove top, arranging the 4 control knobs in a 2x2 grid corresponding exactly to the 2x2 layout of the 4 gas burners provides natural mapping.

Q3: Write names of six usability attributes.

1. Learnability
2. Efficiency
3. Memorability
4. Errors (Low error rate & recovery)
5. Effectiveness
6. Satisfaction

Q4: What is the difference between perceived affordance and real affordance?

Real affordance refers to the actual physical properties of an object that allow a specific action (e.g., a physical button can be physically pressed down). Perceived affordance refers to the visual cues provided by the design that communicate to the user how an object can be interacted with (e.g., 3D drop-shadows on a digital screen button making it look pressable).

Q5: A prototype drawn on paper by hand is an example of low fidelity or high fidelity? Justify shortly.

It is a Low-Fidelity (Lo-Fi) prototype. Justification: It has low visual and functional resemblance to the finished software product, uses primitive non-digital media (paper/pencil), and is designed for rapid, low-cost exploration of concepts rather than detailed interaction design.

Q6: Can we have medium-fidelity prototypes? Explain shortly.

Yes. Medium-fidelity prototypes combine elements of both lo-fi and hi-fi design. They typically feature wireframes built using digital tools (like Balsamiq) with clickable screen transitions and structured layout, but lack final aesthetic polish, full data persistence, or realistic visual styling.

Q7: State four things provided to evaluators during a briefing session in Heuristic Evaluation.

1. A checklist of usability heuristics (e.g., Nielsen's 10 Heuristics).
2. Description of the system prototype or interface mockup.
3. Target user descriptions / Personas.
4. Representative task descriptions / Scenarios.

Q8: Differentiate between throw-away and evolutionary prototypes.

Throw-away prototypes are quick, temporary models created solely to gather requirements and test ideas, after which they are completely discarded. Evolutionary prototypes are structured models that are continuously refined and expanded stage-by-stage based on user feedback until they evolve into the final production system.

Q9: What is the core difference between formative and summative evaluation?

Formative evaluation is conducted DURING the development process to inform design improvements and collects qualitative process data (why errors happen). Summative evaluation is conducted AFTER system completion to assess overall performance against benchmarks and collects quantitative bottom-line data (task completion times, error rates).

Q10: Differentiate between recognition and recall.

Recognition involves identifying a correct option using visible visual cues (Knowledge in the World), such as picking a command from a GUI dropdown menu. Recall involves retrieving information from memory without any external assistance (Knowledge in the Head), such as typing Unix commands in a CLI.

Q11: Differentiate between internal and external consistency with examples.

Internal consistency means maintaining identical visual layouts, terminology, and behavior within the same application (e.g., placing the 'Cancel' button to the right of 'OK' across all dialogs in an app). External consistency means matching conventions across different applications on the same platform (e.g., using Ctrl+C for Copy across all Windows software).

Q12: How are inspections different from testing?

Usability Inspections are expert-based evaluations where HCI specialists examine interfaces using heuristics or walkthroughs without involving real end-users. Usability Testing involves observing representative real end-users performing tasks on the prototype to measure performance and capture feedback.

Q13: What is co-experience with respect to primary and secondary users?

Co-experience refers to the social and emotional user experience created when primary users (who interact directly with the app, e.g., a patient using a health app) and secondary users (who receive indirect outputs or provide care, e.g., a doctor reviewing patient logs) interact together through the shared system.

Q14: What are the measurable attributes of a system's rejectability?

1. High Cost (unaffordable).
2. Incompatibility with existing software/hardware.
3. Unreliability and high crash rates.
4. Social or Legal Acceptability violations (violating privacy, laws, or cultural norms).

Q15: Differentiate between novelty and meaningfulness in innovation.

Novelty means an object or process is relatively new to the environment and users. Meaningfulness means the novelty actually solves a real user need, want, or irritation, making it useful and adoptable on a large scale. Innovation = Novelty + Meaningfulness.

Q16: How are storyboards different from sketches?

A sketch is a single static drawing representing an interface screen or physical concept. A storyboard is a sequential series of sketch frames linked chronologically with narrative arrows to illustrate a user's scenario, context, and emotional journey over time.

Q17: Differentiate between pre-attentive and attentive processing.

Pre-attentive processing occurs automatically and rapidly (<200 ms) in the low-level visual system for single visual features (color, size). Attentive processing requires conscious, sequential visual search when targets involve feature conjunctions (e.g., finding a red circle among red squares and blue circles).

Q18: Differentiate between utility and usability.

Utility refers to whether the system provides the functionality needed to accomplish the user's task (is it useful?). Usability refers to how easily, efficiently, and pleasantly users can utilize that functionality (is it usable?).

Q19: Explain why context is important when evaluating usability.

Usability is not an inherent static property of software; it depends on the context of use—defined by the specific user class, their domain goals, physical environment (e.g., outdoor glare), and social setting. An interface usable for an office worker may fail completely for a truck driver on the road.

Q20: An elderly person uses a basic keypad phone instead of a smartphone. Classify the user type and explain.

Classified as a Special / Conservative / Extreme User Class (or Non-Elastic User). Reasoning: They prioritize high physical affordances (tactile buttons), low cognitive load, and simplicity over feature richness, avoiding touchscreens due to motor/visual impairments or mental model friction.

Part B: Scenario-Based Past Paper Questions & Solutions

Past Paper Scenario 1: Tesla New Features Innovation Classification

Tesla introduced 4 features: 1. Parental control for cars. 2. YouTube/Netflix in interface. 3. 360-degree camera. 4. Restricted driving profiles (no sports mode under 22 or over 60). Classify each as Incremental or Radical innovation:

- 1. Parental Control for Cars:** *Incremental Innovation.* Justification: It extends existing software restriction algorithms and remote access features already present in modern connected vehicles.
- 2. YouTube/Netflix in Interface:** *Incremental Innovation.* Justification: It embeds existing web streaming media applications into the vehicle's infotainment screen.
- 3. 360-Degree Camera Coverage:** *Incremental Innovation.* Justification: It improves upon existing parking radar/camera systems by synthesizing multi-camera feeds into a seamless view.
- 4. Restricted Driving Profiles (Age/Speed bounds):** *Incremental Innovation.* Justification: It applies standard software profile constraints to vehicle performance parameters.

Past Paper Scenario 2: Soap Tablets Case Study

State whether soap tablets shown in the figure are an innovation or not? Justify your answer. If innovation, state its type.

Solution: Yes, soap tablets are an **Innovation**.

- **Justification:** It possesses both *Novelty* (pre-measured single-use solid tablet replacing liquid soap bottles) and *Meaningfulness* (solves travel liquid limits, eliminates messy leaks, and reduces plastic waste).
- **Type:** *Incremental Innovation*, because it builds upon existing soap chemical formulations and packaging knowledge rather than introducing a radically new cleaning paradigm.

Past Paper Scenario 3: Automated University HR Leave & Attendance System

Pakistani universities automating leave applications (employee applies, boss approves/disapproves, HR manages payroll).

1. User Classes:

- **Primary Users:** Faculty / Staff Employees (apply for leaves) & Immediate Managers / HoDs (approve/disapprove leaves).
- **Secondary Users:** HR Department Officers (manage quotas & payroll) & System Administrators.

2. Persona (Primary User - Faculty):

- **Name:** Dr. Kamran (42, Associate Professor).
- **Goals:** Quick leave submission without manual paper movement; wants real-time status visibility.
- **Frustrations:** Paper leave files get misplaced between HoD office and HR, delaying urgent medical leaves.

3. User Scenario:

- Dr. Kamran wakes up with fever. He opens the University HR portal on his phone, selects 'Casual Leave', selects date range, attaches doctor's note, and clicks 'Submit'. His HoD receives a push notification, reviews the schedule, and clicks

'Approve'. Dr. Kamran immediately gets an automated SMS confirmation that his leave is approved and quota updated.

Part C: Solved Quantitative KLM & Fitts's Law Problems

Problem 1: University Leave Application Form KLM Calculation (Past Paper Q6)

User fills Leave Application Form: Selects Leave Type from dropdown (using mouse), enters 'Date From' (01-07-2024) and 'Date To' (02-07-2024) using keyboard, leaves Remarks blank, uploads supporting file using mouse. Hand initially on keyboard. User is intermediate. Calculate total execution time.

Step-by-Step Operator Table:

Step Description	Operator Sequence	Time (s)
1. Mental prep + Move hand to mouse	M + H	$1.2 + 0.4 = 1.60$ s
2. Point to Leave Type dropdown + Click	P + 2B	$1.1 + 0.2 = 1.30$ s
3. Mental search + Point to 'Casual' + Click	M + P + 2B	$1.2 + 1.1 + 0.2 = 2.50$ s
4. Point to 'Date From' box + Click	P + 2B	$1.1 + 0.2 = 1.30$ s
5. Move hand to keyboard + Mental prep to type	H + M	$0.4 + 1.2 = 1.60$ s
6. Type '01-07-2024' (10 keystrokes)	10K	$10 \times 0.28 = 2.80$ s
7. Move hand to mouse + Point to 'Date To' box + Click	H + P + 2B	$0.4 + 1.1 + 0.2 = 1.70$ s
8. Move hand to keyboard + Mental prep to type	H + M	$0.4 + 1.2 = 1.60$ s
9. Type '02-07-2024' (10 keystrokes)	10K	$10 \times 0.28 = 2.80$ s
10. Move hand to mouse + Point to 'Upload Application' + Click	H + P + 2B	$0.4 + 1.1 + 0.2 = 1.70$ s
11. Mental prep to verify file upload	M	1.20 s
TOTAL EXECUTION TIME	6M + 5P + 5H + 20K + 10B	21.70 Seconds

Part D: Solved Past Paper MCQs Answer Key

#	Past Paper MCQ Question	Correct Answer	HCI Concept & Rationale
1	Four main components for an interaction to happen?	User, Computer, Input, Output (or S, O, I, U)	Abowd & Beale interaction framework.
2	Factors in physical environment affecting interaction quality?	All of the above (Noise, Temp, Lighting)	Ergonomics & environmental constraints.
3	For a software product to be successful, it must be:	All of the above (Useful, Usable, Used)	Utility + Usability = Practical Acceptability.
4	Which statement is true regarding personas?	A primary persona needs their own interface	Primary personas drive the core UI model.
5	Refers to how good a system is at doing what it is supposed to do:	Usefulness	System Acceptability framework (Slide 02).
6	Macintosh menu bar is faster than Windows menu bar because:	Screen edges provide infinite target width (W)	Fitts's Law Mile-High Menu principle.
7	Why cascading submenus are poor design?	They require steering (difficulty is linear D/S)	Steering Law formula $T = a + b(D/S)$.
8	Standard time for Mental operator (M) in KLM?	1.2 seconds	Standard KLM operator value.
9	MHP Cognitive Processor cycle time (T_c)?	70 ms	Model Human Processor standard.
10	Severity rating 4 in Heuristic Evaluation means:	Usability catastrophe	Imperative to fix before release.